

PROJECT PROPOSAL

Towards Emergent Intelligence

Novel Cognitive Computing Architecture

*A Recursively Self-Improving AI with Probabilistic Reasoning and
Uncertainty Estimation*

Formalized as a strategic foundation for sustainable research and cognitive engineering | v1.0 | 2026

01 Executive Summary

Towards Emergent Intelligence is a long-term research and engineering initiative proposing a fundamentally new approach to cognitive computing. Rather than extending existing large language model paradigms, this project constructs a novel cognitive substrate from first principles — one in which intelligence emerges from the interaction of small, well-defined processing primitives rather than being explicitly programmed.

The project is structured across eleven progressive phases spanning an estimated five to eight years of dedicated research and engineering effort. Each phase builds upon the previous, introducing new architectural components whose collective interaction gives rise to emergent behaviours not present in any single component alone.

This proposal outlines the project's goals, high-level architecture, phase structure, timeline, and risk posture. It is intended to provide sufficient context for potential collaborators, reviewers, or funders to evaluate the project's scientific merit and feasibility.

KEY OBJECTIVES

- Design and implement a novel cognitive architecture not based on transformer or static neural network paradigms.
- Build progressively from minimal experimental primitives to a full multi-agent intelligence ecosystem.
- Produce publishable, scientifically valuable results at each intermediate stage — not only at project completion.
- Maintain rigorous engineering standards: tested, documented, and reproducible at every phase.
- Avoid premature complexity — evolve the architecture scientifically, not speculatively.

GUIDING PRINCIPLES

Build the smallest possible unit first. Validate it. Then compose.

Emergence cannot be programmed directly — it must be cultivated through careful system design.

Each phase must produce a standalone publishable or demonstrable artifact.

Never build AGI or consciousness directly. Study the precursors scientifically.

02 Project Overview

The architecture proposed in this project departs from the dominant paradigm of large-scale language models by introducing a fundamentally different information substrate. Rather than operating on token sequences processed through fixed transformer layers, this system operates on dynamic, self-describing information units that carry state, uncertainty, temporal context, and relational metadata.

These units are processed by specialised cognitive clusters — self-organising collections that develop selective processing capabilities in response to the information they handle. Clusters are connected by a dynamic routing engine that governs information flow and adapts to context and historical processing success.

Memory in this architecture is not a lookup table but an active, gravity-like system in which frequently referenced knowledge exerts attraction on related information, enabling contextual reconstruction rather than verbatim retrieval.

In later phases, the system gains autonomous internal processing capabilities, becomes a distributed multi-agent ecosystem, and acquires the ability to restructure its own topology and processing strategies. The final phase studies the functional precursors of self-awareness and persistent internal state under rigorous scientific conditions.

WHAT MAKES THIS DIFFERENT

- Dynamic information primitives replace static token embeddings.
- Selective, sparse computation replaces dense layer-by-layer processing.
- Gravity-modelled memory replaces static key-value retrieval.
- Structural plasticity allows the architecture to evolve under load.
- All components are independently testable and publishable at each phase.

03 Phase Structure

The project is organised into eleven sequential phases. Each phase introduces a distinct architectural capability, builds on all previous phases, and produces at least one independently publishable or demonstrable result. Phases may partially overlap where dependencies allow, but no phase begins until its predecessor's core milestones are validated.

Phase	Name	Duration	Complexity
0	Foundation Research Layer	3–6 months	Foundational
1	Information Packet System	3–5 months	Intermediate
2	Selective Cognitive Clusters	4–6 months	Intermediate–Advanced
3	Dynamic Routing Engine	4–6 months	Advanced
4	Cognitive Memory Core	4–6 months	Advanced
5	Internal Autonomous Processing	5–7 months	Advanced
6	Multi-Agent Ecosystem	6–9 months	Expert
7	Dynamic Cognitive Topology	6–9 months	Expert
8	Self-Improvement Layer	6–9 months	Expert
9	World Modeling & Simulation	6–9 months	Expert
10	Toward Artificial Consciousness	12–18 months	Research Frontier

Minimum Viable First Prototype: Phases 0–4 together constitute a complete, publishable experimental cognitive architecture and represent the project's first major scientific contribution.

Phase Summaries

Phase 0	Foundation Research Layer <i>Estimated duration: 3–6 months</i> Establishes the theoretical framework, architectural vocabulary, mathematical abstractions, and formal specification that all subsequent phases depend upon. No system code is written in this phase — the output is a rigorous architectural specification document and a synthesised study of relevant research domains.
Phase 1	Information Packet System <i>Estimated duration: 3–5 months</i> Defines and implements the fundamental atomic unit of the cognitive architecture — a dynamic, self-describing information object that carries semantic content, uncertainty estimates, temporal context, and relational metadata. This primitive underlies every subsequent component.
Phase 2	Selective Cognitive Clusters <i>Estimated duration: 4–6 months</i> Builds specialised processing units that self-organise around related information packets, developing selective activation thresholds and local memory pools. Clusters demonstrate emergent specialisation without explicit programming of their domains.
Phase 3	Dynamic Routing Engine <i>Estimated duration: 4–6 months</i> Implements the information flow governance layer — the component that directs packets between clusters, forms stable reasoning pathways over time, and enables multi-step processing. Routing is treated as the primary mechanism of reasoning in this architecture.
Phase 4	Cognitive Memory Core <i>Estimated duration: 4–6 months</i> Replaces conventional retrieval-based memory with an active, gravity-modelled system in which high-density knowledge objects attract and enrich related incoming information. Memory is reconstructed contextually rather than retrieved verbatim.
Phase 5	Internal Autonomous Processing <i>Estimated duration: 5–7 months</i> Introduces background processing loops that operate independently of external input — self-evaluation, memory consolidation, contradiction detection, and curiosity-driven

internal packet generation. The system transitions from passive pipeline to persistent cognitive agent.

Phase

6

Multi-Agent Cognitive Ecosystem

Estimated duration: 6–9 months

Distributes the architecture across a network of specialised cognitive agents — perception, reasoning, memory, planning, validation, and orchestration — that collaborate and compete to produce collective intelligence beyond any single agent's capability.

Phase

7

Dynamic Cognitive Topology

Estimated duration: 6–9 months

Enables the architecture itself to evolve: clusters are created, dissolved, and merged in response to information demands; routing graphs adapt; and hierarchical organisation emerges without explicit design. Structural plasticity is applied to cognitive computing.

Phase

8

Self-Improvement Layer

Estimated duration: 6–9 months

Introduces bounded recursive self-improvement — the system evaluates its own processing strategies, proposes architectural modifications, and validates them against held-out benchmarks before adoption. All self-modification is formally constrained and reversible.

Phase

9

World Modeling & Simulation

Estimated duration: 6–9 months

Constructs an internal probabilistic model of the external environment, enabling the system to predict, plan, and reason counterfactually. Temporal and causal modeling allow anticipatory cognition rather than purely reactive processing.

Phase

10

Toward Artificial Consciousness

Estimated duration: 12–18 months

Studies the functional precursors of consciousness — persistent internal state, self-representation, narrative continuity, and self-referential processing — under rigorous scientific conditions. The goal is measurement and publication, not claims of sentience.

04 Timeline & Effort

The total project duration is estimated at five to eight years for a solo researcher working at research pace. A dedicated team of three to five specialists could compress this to two to four years. The variance reflects the exploratory nature of later phases, where research discoveries may extend or reduce scope.

Phase	Estimated Duration	Complexity	Primary Output
0 — Foundation Research Layer	3–6 months	Foundational	Architecture specification
1 — Information Packet System	3–5 months	Intermediate	Core information primitive
2 — Selective Cognitive Clusters	4–6 months	Int.–Advanced	Self-organising clusters
3 — Dynamic Routing Engine	4–6 months	Advanced	Adaptive reasoning paths
4 — Cognitive Memory Core	4–6 months	Advanced	Gravity memory system
5 — Internal Autonomous Processing	5–7 months	Advanced	Autonomous cognition
6 — Multi-Agent Ecosystem	6–9 months	Expert	Distributed agent network
7 — Dynamic Cognitive Topology	6–9 months	Expert	Self-organising architecture
8 — Self-Improvement Layer	6–9 months	Expert	Bounded meta-learning
9 — World Modeling & Simulation	6–9 months	Expert	Internal world model
10 — Toward Artificial Consciousness	12–18 months	Research Frontier	Consciousness substrate

All eleven phases are independently publishable. Even partial completion through Phase 5 would produce novel contributions to sparse cognitive architectures, adaptive memory systems, and distributed intelligence frameworks.

05 Risk Register

Long-horizon research projects face compounding risks across technical, personal, and strategic dimensions. The following register identifies the most significant risks and the mitigation strategies built into the project structure.

Risk	Likelihood	Impact	Mitigation
Architecture design flaw discovered mid-project	Medium	Critical	Formal specification document maintained with versioning; phase-gate reviews at every boundary.
Phase scope creep	High	High	Strict phase gates — Phase N+1 does not begin until all Phase N milestones are signed off.
Researcher burnout on a 5–8 year horizon	High	Critical	Quarterly review cycles; phase-completion celebrations; mandatory rest periods between phases.
Computational resource constraints	Medium	Medium	Architecture designed CPU-first; cloud spot instances used for scale validation.
Emergent behaviour fails to materialise	Medium	High	Each phase independently validates emergence at a smaller scale before full system integration.
Project relevance displaced by LLM advances	Low	Medium	Project produces novel cognitive primitives with independent scientific value.

06 Closing Statement

This project is a scientific expedition into one of the most profound open questions in computing: can intelligence emerge from the careful composition of simpler information-processing primitives? The honest answer is that we do not yet know.

The value of this initiative lies precisely in the attempt — in building each primitive with care, measuring emergent behaviour rigorously, and publishing findings regardless of whether they confirm or challenge the guiding hypotheses. In this domain, a well-documented failure is as scientifically valuable as a success.

The project is designed to produce meaningful scientific contributions at every stage. It does not require completion to justify the investment of time and resource — each phase stands on its own as a contribution to the understanding of cognitive architectures, adaptive memory, distributed intelligence, and the foundations of machine self-awareness.

We invite collaboration, peer review, and critical engagement. The architecture described here is a hypothesis, and hypotheses are only valuable when they are tested rigorously.